VANISHING OF NEGATIVE EXT GROUPS AND THE TOP EXT IDENTIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 8946-8960, following the statement at 8929-8944.

Editorial note. The top-Ext identification compresses the standard bookkeeping: with L^i = 0 for i < a and Y^j = 0 for j < b only the (a, b) component survives, modulo boundaries on both sides.

Allowed standard prerequisites (editorial).

- Truncation quasi-isomorphisms. same chapter truncation machinery (used at ~8944 and ~8948). Replaces Y by tau_{>=b}Y and restricts resolutions to tau_{<=a}L.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category.*

(1) *Let X, Y be objects of $D(\mathcal{A})$. Given $a, b \in \mathbf{Z}$ such that $H^i(X) = 0$ for $i > a$ and $H^j(Y) = 0$ for $j < b$, we have $\mathrm{Ext}^n_{\mathcal{A}}(X, Y) = 0$ for $n < b - a$ and*

$$\mathrm{Ext}^{b-a}_{\mathcal{A}}(X, Y) = \mathrm{Hom}_{\mathcal{A}}(H^a(X), H^b(Y))$$

(2) *Let $A, B \in \mathrm{Ob}(\mathcal{A})$. For $i < 0$ we have $\mathrm{Ext}^i_{\mathcal{A}}(B, A) = 0$. We have $\mathrm{Ext}^0_{\mathcal{A}}(B, A) = \mathrm{Hom}_{\mathcal{A}}(B, A)$.*

Proof. Choose complexes $X^\bullet$ and $Y^\bullet$ representing X and Y. Since $Y^\bullet \to \tau_{\geq b} Y^\bullet$ is a quasi-isomorphism, we may assume that $Y^j = 0$ for $j < b$. Let $L^\bullet \to X^\bullet$ be any quasi-isomorphism. Then $\tau_{\leq a} L^\bullet \to X^\bullet$ is a quasi-isomorphism. Hence a morphism $X \to Y[n]$ in $D(\mathcal{A})$ can be represented as fs^{-1} where $s : L^\bullet \to X^\bullet$ is a quasi-isomorphism, $f : L^\bullet \to Y^\bullet[n]$ a morphism, and $L^i = 0$ for $i < a$. Note that f maps L^i to Y^{i+n}. Thus $f = 0$ if $n < b - a$ because always either L^i or Y^{i+n} is zero. If $n = b - a$, then f corresponds exactly to a morphism $H^a(X) \to H^b(Y)$. Part (2) is a special case of (1). □

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